→ Friend function :-

A friend function in C++ is a function that is not a member of a class but has access to its private and protected members.

It is declared using the keyword friend inside the class.

- Useful when two or more classes need to work closely together and access each other's private data.

→ // c program;

```
#include<iostream>
using namespace std;
class student
{
    private:
      int age;
      student(int a) //constructor to initialize age
      {
        age=a;
      }
      //friend function declaration
    friend void show(student);
};
int main()
{
```

Akash

```
// friend function definition

void show(student s1)
{
    cout<<"age= "<<s1.age;
}

int main()
{
    student s(20);
    show(s); // calling friend function
    return 0;
}
```

Advantag:-

→ Access to private data outside the class
→ Also used in operator overloading
→ Useful when two or more classes need to work closely together.

Guideline for using friend function:-

→ Use when necessary because it breaches encapsulation and oops priciple.

Akash

→ Inline function :-

It is a special function in C++ used as an improvement over macro of C language.

The compiler copies the body of function at all the places where it is called so it removes the time requirement due to context switching, so execution will be fast.

It is declared by using the keyword inline inside the class.

→ // C program;

```
#include<iostream>
using namespace std;
class student
  {
      public:
       inline void sum(int a, int b) // inline function
         {
           cout<< "sum= " << a+b <<endl;
         }
  };
     int main()
  {
       Student s;
```

Akash

```
s.sum(20,12); // calling inline function
s.sum(20,04);
s.sum(18,04);
s.sum(20,25);
return 0;
}
```

Akash